BIOCHEMISTRY CUMULATIVE EXAM

October 26, 1991

A given solute S may enter the cell by two methods: passive diffusion across the cell membrane or facilitated transport in which a saturable transport protein moves the solute across the membrane. The rate at which the concentration of S increases within the cell ($v = d[S]_{cell}/dt$) is given by Equation I below. The rate at which the concentration of S increases within the cell due to facilitated transport is given by Equation II below. (For simplicity, we are assuming that the initial concentration of the solute in the cell is 0 and that initial rates only are being considered.)

(I) Passive: $v = k\ [S]_{medium}$

where k is a constant related to the permeability coefficient and the volume and surface area of the cells

(II) Facilitated: $v = Vmax\ [S]_{medium}/(Km + [S]_{medium})$

where Km and Vmax are constants with meanings similar to those encountered in enzyme kinetics

You have collected data on the rate of uptake of the solute S by a suspension of cells and collected the following data.

$[S]_{medium}$ (mM)	$v\ (= d[S]_{cell}/dt)$ mM/min
0.0	0.0
0.1	1.72
0.2	2.95
0.5	5.25
1.0	7.17
2.0	9.00
5.0	11.6
10.0	14.5
20.0	19.8
50.0	34.9

a. (20 points) Sketch a general curve showing how the rate of entry of S into the cell would depend on the concentration of S in the medium if only passive diffusion across the membrane were occurring. Also sketch a general curve as to what you would expect if facilitated transport only were occurring.

b. (20 points) Using the graph paper provided, determine whether the uptake of S by the cell suspension in question involves passive diffusion, facilitated transport, or both? Justify your answer.

c. (60 points) Depending on your answer in (b), determine as many of the constants in Equations I and II as possible. Explain the logic and calculation necessary to arrive at your answers.